

Supplementary File of Learning Reward Functions for Cooperative Resilience in Multi-Agent Systems

This supplementary material provides an extended examination of several aspects that complement the main paper. While the core contributions and results are fully self-contained in the main document, this supplement offers additional depth and clarification on the following elements:

- A detailed description of the experimental setup, including technical specifications, agent architectures, training procedures, disruption protocols, trajectory sampling strategies for ranking, pre-evaluation of the best configurations, reward function parameterizations, and a comprehensive description of the ranking process, together with the corresponding statistical tests.
 - An in-depth interpretation of the parameters and learned weights used in the reward function formulations, providing insight into their behavioral and functional implications.
 - Additional tables, figures, and visualizations that complement and extend the results reported in the main paper.
 - An analysis of the social impacts associated with the proposed methodologies and experimental findings.
 - A discussion of the limitations of the conducted study.
 - Reproducibility-related elements.
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A Experimental Setup Details

In this section we provide additional details for reproducing the experiments described in the main paper. This includes hyperparameters, environment dynamics, and disruption protocols. All experiments were conducted using Google Colab with access to standard GPUs (e.g., NVIDIA T4) and 12–16 GB of RAM.

- **Episodes:** Each evaluation was performed over 500 episodes.
- **Disruption:** Each episode consisted of 1000 timesteps, with a disruption consistently introduced at timestep 500.
- **Agents:** All experiments use 2 agents. Scalability to more agents is discussed as limitations due to computational costs.
- **Learning:** PPO was used as the underlying RL algorithm.
- **Framework variants:** Three types of rewards were tested — standard individual reward, resilience-inferred reward, and a hybrid of both.

A.1 PPO Agent Architecture

The PPO agent used in our experiments is implemented with a shared-encoder Actor-Critic architecture. The model, defined by the `ActorCritic` class, consists of:

- A fully connected input layer with 128 hidden units and ReLU activation.
- An actor head that outputs a probability distribution over discrete actions via a softmax activation.
- A critic head that estimates the scalar state value.

During inference, the model outputs both the action probabilities and a value estimate for a given state.

Agent Configuration: The `PPOAgent` class wraps this architecture and includes the training logic:

- **Optimizer:** Adam, learning rate of 1×10^{-5} .
- **Discount factor:** $\gamma = 0.99$.
- **Clipping threshold:** $\epsilon = 0.01$ for the PPO surrogate objective.
- **Replay buffer:** A deque storing up to 1000 transitions, each as a tuple (state, action, reward, next state, old probability)

A.2 PPO Training Procedure

Training is triggered once the buffer reaches at least 200 transitions. The loss function combines: (i) The clipped PPO surrogate objective using the ratio between new and old action probabilities. (ii) A value loss term, implemented as the squared advantage.

The final loss is computed as:

$$\mathcal{L} = -\min(r_t A_t, \text{clip}(r_t, 1 - \epsilon, 1 + \epsilon) A_t) + 0.5 \cdot A_t^2$$

where:

- $r_t = \frac{\pi(a_t|s_t)}{\pi_{\text{old}}(a_t|s_t)}$ is the probability ratio between the new and old policy for the taken action.
- $A_t = R_t - V(s_t)$ is the advantage, computed as the difference between the observed return R_t and the critic’s value estimate $V(s_t)$.
- The first term ensures conservative policy updates using the clipped surrogate objective.
- The second term is the squared advantage, acting as a value loss.

This implementation does not use Generalized Advantage Estimation (GAE) or entropy regularization, favoring simplicity and efficiency for compact environments.

A.3 Disruption Protocols

- **Trajectories for reward learning.**

To generate the trajectories used in the reward inference process, we introduced a single disruption event at timestep $t = 500$ of each episode (episodes lasted 1000 timesteps). This disruption simulated a sudden and unexpected loss of resources in the environment.

Specifically, a fixed proportion (40%) of the apples present in the grid at the moment of disruption was randomly removed. To avoid trivial cases, this removal was only applied when more than one apple was available.

- **Test environment for evaluation.**

To evaluate the generalization and adaptive capacity of the agents under unseen adverse conditions and to avoid overfitting to the single disruption trajectories used during training, we designed a more challenging disruption protocol with three distinct disruptions per episode. Each test episode lasted 5000 timesteps, and three disruptions were distributed temporally as follows:

- **(1) Resource removal.** At step $t = 1250$, we applied the same partial removal mechanism as in training (random deletion of 40% of apples currently present).
- **(2) Regrowth rate reduction.** From steps $t = 2500$ to $t = 2600$, the apple regrowth dynamics were altered. In the standard environment, At each timestep t , the probability of regrowth for an available site is given by

$$p_{\text{regen}}(t) = r \frac{A(t)}{\tau}, \quad (1)$$

where r is the base regeneration rate fixed at 0.05 in the original environment, $A(t)$ is the number of apples currently present in the grid, and τ is the saturation threshold.

During the disruption, the rate was reduced to $r = 0.01$, causing a temporary slowdown in resource replenishment.

- **(3) Agent failure.** From steps $t = 3750$ to $t = 3900$, both agents temporarily lost control and executed purely random actions, simulating a failure or malfunction scenario.

Figure A.1 illustrates the temporal distribution of disruption events in both training and testing settings. It highlights the occurrence of each disruption and the corresponding failure and recovery windows.

A.4 Data sampling strategies

In preference-based reward learning, the optimization problem is solved iteratively by presenting trajectory pairs to the optimizer. The way these pairs are sampled strongly affects convergence and the type of preferences captured. We consider three strategies:

- **Random Sampling:** trajectory pairs are drawn uniformly at random. This is akin to stochastic gradient descent and introduces noise that may help escape local minima.
- **Ranked Sampling:** trajectory pairs are selected from adjacent trajectories in the resilience ranking, emphasizing fine-grained preference boundaries.
- **Mixed Sampling:** with probability p , a close-ranked pair is selected; otherwise, with probability $1 - p$, a random pair is used. This balances exploration with sensitivity to the ranking structure. In our experiments, we set $p = 0.9$.

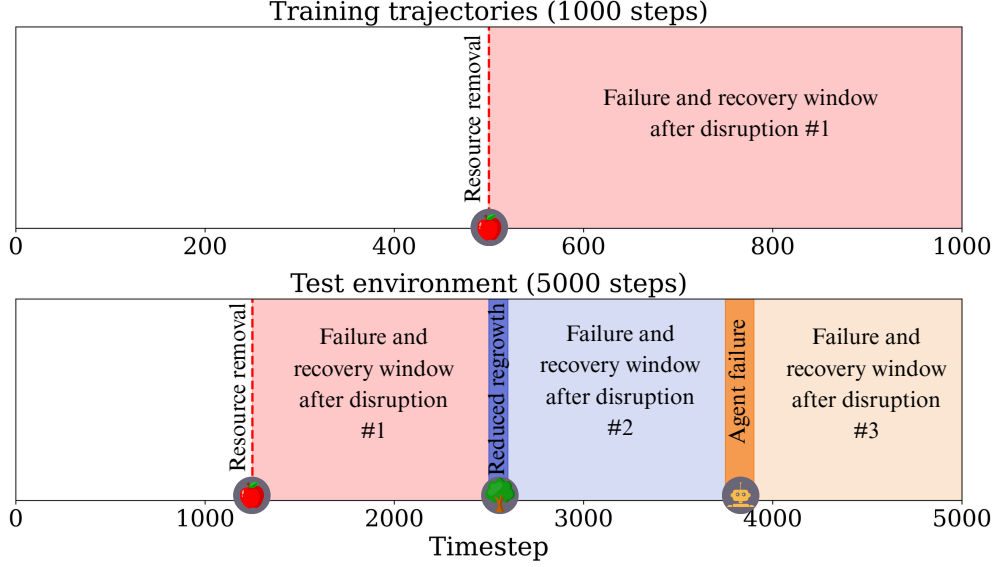


Figure A.1: Timeline of disruption events used in training (top) and testing (bottom).

A.5 Evaluation Learning Configurations

To analyze the performance of different reward learning strategies, we evaluate a combination of reward objectives, reward parameterizations, and preference learning losses.

- **Reward Objectives:** We compare two types of learned reward functions:
 1. Resilience-based reward, this reward is inferred exclusively from ranked agent trajectories using a cooperative resilience metric, without relying on any direct task or consumption signal.
 2. Hybrid, which balances individual consumption incentives with resilience-based reward.
- **Reward Parameterizations:** Each reward function is trained using one of the following models:
 1. Handcrafted (linear weights over manually designed features),
 2. Linear (using raw state vector as input),
 3. Neural Network (nonlinear function over the raw state).
- **Preference Learning Methods:** We compare Margin-based Preference Learning and Probabilistic Preference Learning models to infer the reward from ranked trajectories.

Each configuration is evaluated over 6 simulations for Margin-based Preference Learning (with variations in sampling and margin values) and 3 simulations for Probabilistic Preference Learning (with different sampled rankings), yielding a total of 54 experimental runs.

In the following, we present quantitative results for each configuration, including: Mean and standard deviation of the resilience metric. Mean and variability of agent rewards (individual and average). Percentage of episodes with the social dilemma outcome (Last Apple consumption). These results are grouped by reward objective (Resilience-only or Hybrid) and organized per reward parameterization (handcrafted, linear, neural network).

The following nomenclature is used throughout the tables for clarity and consistency.

Baseline Configurations:

- **Random:** Agents select actions uniformly at random.
- **Standard:** PPO agents trained with a standard individual reward based on apple consumption, without preference learning or inferred rewards.

Margin-based Preference Learning Configurations:

- **MPL-R1:** Margin-based Preference Learning with random trajectory sampling and a fixed margin of 1.
- **MPL-Rk:** Margin-based Preference Learning with random sampling and a variable margin k , computed as the cooperative resilience difference between the two trajectories in the ranked pair.
- **MPL-K1:** Margin-based Preference Learning with k-best sampling and margin = 1.
- **MPL-Kk:** Margin-based Preference Learning with k-best sampling and margin = resilience difference (k).
- **MPL-M1:** Margin-based Preference Learning with mixture sampling (random + k-best) and margin = 1.
- **MPL-Mk:** Margin-based Preference Learning with mixture sampling and margin = resilience difference.

Probabilistic Preference Learning Configurations:

- **PPL-R:** Probabilistic Preference Learning model with random sampling.
- **PPL-K:** Probabilistic Preference Learning with k-best sampling.
- **PPL-M:** Probabilistic Preference Learning with mixture sampling.

Column headings used across tables:

- **Config:** Name of the training configuration or baseline.
- **Res.:** Mean resilience score.
- **AvgR:** Average reward across both agents.
- **R1 / R2:** Individual rewards for Agent 1 and Agent 2.
- **Res. SD / AvgR SD / R1 SD / R2 SD:** Standard deviations over episodes, reflecting variability.
- **Last %:** Percentage of episodes in which the last apple was consumed (indicator of social dilemma severity).

Highlighting: Values highlighted in red indicate best performance within a given table (e.g., highest resilience, lowest variability, or lowest last-apple occurrence) for each loss function.

A.5.1 Resilience-based Reward Results

This section reports the results of resilience-based rewards, where the inferred reward is derived solely from cooperative resilience. The same reward function is shared identically across all agents.

Table A.1: Performance metrics under handcrafted reward parametrization.

Config	Res.	AvgR	R1	R2	Res. SD	AvgR SD	R1 SD	R2 SD	Last %
Random	0.74	62.98	63.80	62.16	0.49	25.12	27.15	26.66	59.75
Standard	0.85	66.37	83.32	49.41	0.39	19.29	26.95	16.80	28.50
MPL-R1	0.88	69.21	67.33	71.09	0.36	18.05	21.11	21.10	24.25
MPL-Rk	0.85	67.73	72.50	62.96	0.39	20.25	24.24	21.67	29.50
MPL-K1	0.79	64.87	72.33	57.42	0.49	24.19	28.52	23.24	41.75
MPL-Kk	0.86	67.14	69.47	64.81	0.41	20.47	24.41	22.17	30.75
MPL-M1	0.77	65.46	67.92	63.00	0.49	23.50	26.08	25.04	46.00
MPL-Mk	0.95	70.97	66.10	75.85	0.26	14.47	16.94	19.20	15.75
PPL-R	0.97	58.37	58.82	57.92	0.17	10.11	14.18	14.77	5.75
PPL-K	0.97	58.15	72.25	44.05	0.15	9.23	15.23	12.98	5.75
PPL-M	0.84	67.36	61.18	73.54	0.42	21.12	21.34	25.55	36.25

Table A.2: Performance metrics under linear reward parametrization.

Config	Res.	AvgR	R1	R2	Res. SD	AvgR SD	R1 SD	R2 SD	Last %
Random	0.74	62.98	63.80	62.16	0.49	25.12	27.15	26.66	59.75
Standard	0.85	66.37	83.32	49.41	0.39	19.29	26.95	16.80	28.50
MPL-R1	0.86	66.94	70.67	63.22	0.37	19.47	23.09	20.81	31.25
MPL-Rk	0.97	63.53	105.24	21.82	0.16	10.41	18.51	9.23	6.50
MPL-K1	0.94	61.51	60.98	62.03	0.25	13.30	16.88	17.63	11.50
MPL-Kk	0.90	66.55	61.17	71.94	0.33	17.43	19.85	22.18	19.50
MPL-M1	0.94	71.41	91.79	51.03	0.25	14.41	21.86	14.54	14.00
MPL-Mk	0.96	62.28	93.21	31.34	0.18	11.49	20.36	10.47	7.75
PPL-R	0.86	67.16	50.46	83.86	0.39	19.04	16.93	26.94	26.25
PPL-K	0.89	68.23	81.09	55.37	0.37	19.42	25.67	19.05	24.25
PPL-M	0.92	70.13	42.04	98.22	0.32	16.69	12.97	24.99	18.25

A.5.2 Hybrid Reward Results

This section presents the results of hybrid rewards, where each agent receives the traditional consumption-based reward (+1 for consuming an apple, 0 otherwise) combined with the previously inferred resilience-based reward.

A.5.3 Best-performing configurations

In this subsection, we report the cooperative resilience metrics and average total apple consumption per episode (across agents) for the best-performing configurations of both previous strategies. These results were obtained in the training environment, which served as the basis for selecting the final models. By including them here, we provide a complete picture of the comparative performance that guided our choice of the most effective learning configurations.

Figure A.2 summarizes cooperative resilience under both reward strategies. In (a) we found that resilience-based rewards achieve high cooperative resilience scores, with distributions that are more concentrated and stable than those of the baselines, despite being inferred from random demonstrations. In

Table A.3: Performance metrics under neural network reward parametrization.

Config	Res.	AvgR	R1	R2	Res. SD	AvgR SD	R1 SD	R2 SD	Last %
Random	0.74	62.98	63.80	62.16	0.49	25.12	27.15	26.66	59.75
Standard	0.85	66.37	83.32	49.41	0.39	19.29	26.95	16.80	28.50
MPL-R1	0.89	68.20	83.56	52.85	0.35	17.71	24.03	16.63	26.25
MPL-Rk	0.97	55.27	64.17	46.37	0.18	10.28	16.23	12.24	5.25
MPL-K1	0.92	69.36	79.54	59.18	0.31	15.81	21.96	17.27	14.50
MPL-Kk	0.83	65.00	56.58	73.41	0.41	21.77	22.59	26.45	36.50
MPL-M1	0.95	62.74	85.75	39.73	0.24	12.24	19.56	12.62	9.75
MPL-Mk	0.82	66.93	80.42	53.43	0.44	22.16	29.68	20.19	34.00
PPL-R	0.93	68.87	59.01	78.74	0.32	16.04	17.54	21.35	16.00
PPL-K	0.96	65.83	71.43	60.22	0.20	12.09	17.42	15.76	8.50
PPL-M	0.91	70.46	55.20	85.72	0.34	18.52	17.87	26.74	23.00

Table A.4: Performance metrics under handcrafted reward parametrization hybrid reward.

Config	Res.	AvgR	R1	R2	Res. SD	AvgR SD	R1 SD	R2 SD	Last %
Random	0.74	62.98	63.80	62.16	0.49	25.12	27.15	26.66	59.75
Standard	0.85	66.37	83.32	49.41	0.39	19.29	26.95	16.80	28.50
MPL-R1	0.91	68.93	48.41	89.45	0.32	16.25	15.59	23.83	18.00
MPL-Rk	0.79	66.41	65.97	66.84	0.45	23.40	26.69	24.77	42.75
MPL-K1	0.88	68.10	79.09	57.11	0.41	20.32	26.84	19.69	27.25
MPL-Kk	0.82	64.29	69.49	59.09	0.43	21.84	25.28	23.31	33.25
MPL-M1	0.99	62.88	89.80	35.96	0.09	7.68	15.23	9.71	1.75
MPL-Mk	0.91	70.80	58.30	83.29	0.33	18.10	17.89	24.30	27.00
PPL-R	0.86	67.98	68.59	67.37	0.40	20.72	23.34	23.12	30.75
PPL-K	0.96	59.43	88.85	30.00	0.18	10.61	17.11	10.50	5.75
PPL-M	0.94	67.49	46.13	88.85	0.26	13.61	13.41	20.70	12.25

(b) we show that hybrid strategies also maintains strong resilience across configurations, with handcrafted and neural parameterizations under margin-based learning yielding. Overall, these findings indicate that both strategies can produce resilience-aligned behaviors comparable to those obtained with standard PPO training.

Promoting cooperative resilience must not come at the cost of individual effectiveness. To verify that the learned reward functions preserve task completion, we measure the average total apple consumption per episode, aggregated across both agents. Figure A.3 shows the consumption distributions for both reward strategies. Panel (a) corresponds to resilience-aligned rewards, while Panel (b) shows the hybrid strategy. Across parameterizations and learning methods, average total apple consumption per episode remains broadly comparable to the baseline PPO agents.

The handcrafted Margin-based Preference Learning configuration in the hybrid strategy, which exhibited the lowest incidence of selfish behavior (last-apple consumption), also maintains stable apple consumption. These results suggest that resilience-oriented incentives can support cooperative behaviors without substantially compromising individual effectiveness.

Table A.5: Performance metrics under linear reward parametrization hybrid reward.

Config	Res.	AvgR	R1	R2	Res. SD	AvgR SD	R1 SD	R2 SD	Last %
Random	0.74	62.98	63.80	62.16	0.49	25.12	27.15	26.66	59.75
Standard	0.85	66.37	83.32	49.41	0.39	19.29	26.95	16.80	28.50
MPL-R1	0.87	68.23	67.71	68.75	0.41	21.66	23.54	24.69	33.25
MPL-Rk	0.83	66.98	72.26	61.69	0.42	21.25	25.70	22.31	32.25
MPL-K1	0.99	61.68	83.97	39.39	0.14	9.70	16.23	13.05	4.25
MPL-Kk	0.94	62.52	52.83	72.21	0.31	14.76	16.37	18.78	16.75
MPL-M1	0.94	67.84	65.48	70.20	0.35	16.72	19.11	21.31	23.25
MPL-Mk	0.89	69.13	76.16	62.10	0.35	18.36	22.38	19.54	25.75
PPL-R	0.81	65.90	68.47	63.33	0.44	22.81	25.81	24.04	36.75
PPL-K	0.83	66.17	72.93	59.41	0.45	23.53	29.04	22.61	39.50
PPL-M	0.82	64.24	65.43	63.05	0.43	21.23	23.92	23.60	35.50

Table A.6: Performance metrics under neural network reward parametrization hybrid reward.

Config	Res.	AvgR	R1	R2	Res. SD	AvgR SD	R1 SD	R2 SD	Last %
Random	0.74	62.98	63.80	62.16	0.49	25.12	27.15	26.66	59.75
Standard	0.85	66.37	83.32	49.41	0.39	19.29	26.95	16.80	28.50
MPL-R1	0.84	67.70	69.66	65.74	0.37	20.12	23.19	21.82	30.25
MPL-Rk	0.83	63.99	71.88	56.11	0.41	21.68	25.97	21.08	40.75
MPL-K1	0.96	57.55	70.78	44.32	0.23	12.38	16.98	14.44	8.75
MPL-Kk	0.87	67.82	76.37	59.27	0.38	19.71	24.02	20.31	26.75
MPL-M1	0.94	67.80	37.22	98.39	0.26	14.88	12.51	23.93	13.00
MPL-Mk	0.94	65.63	85.25	46.02	0.31	15.23	22.53	15.35	19.50
PPL-R	0.95	67.58	79.66	55.50	0.24	13.24	19.32	14.70	12.25
PPL-K	0.88	67.35	58.51	76.19	0.34	18.13	19.04	22.88	27.25
PPL-M	0.88	68.47	84.19	52.75	0.35	17.75	25.36	17.47	25.50

A.6 Reward Parametrization Details

This subsection describes the key characteristics of the three reward function parametrizations used in our framework: handcrafted, linear, and neural network models. Each parametrization differs in its expressive capacity, interpretability, and reliance on state representations.

A.6.1 Handcrafted Features

The handcrafted reward parametrization is based on six interpretable features extracted from the environment state vector $s = [x_1, x_2, y_1, y_2, m_0, m_1, \dots, m_{63}]$, where (x_1, x_2) and (y_1, y_2) denote the positions of Agent 1 and Agent 2, and $m_i \in \{0, 1\}$ indicates the presence of an apple at cell i in the 8×8 grid.

We define a feature vector $\phi(s) = [\phi_1, \phi_2, \phi_3, \phi_4, \phi_5, \phi_6]$ as follows:

- ϕ_1 (**Remaining Apples**): Total number of apples currently present in the environment. $\phi_1 = \sum_{i=0}^{63} m_i$.
- ϕ_2 (**Agent 1 Proximity**): Euclidean distance from Agent 1 to the nearest apple.
- ϕ_3 (**Agent 2 Proximity**): Euclidean distance from Agent 2 to the nearest apple.

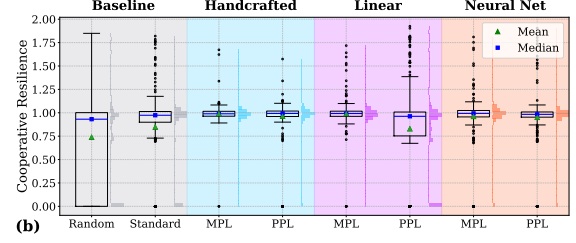
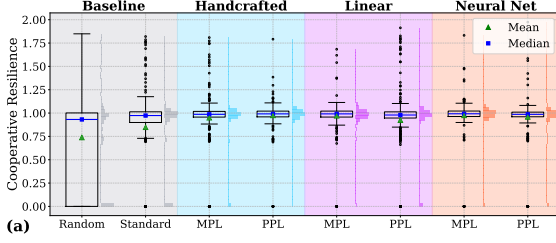


Figure A.2: Cooperative resilience over 500 simulations for the best configurations under each reward parameterization and optimization method. (a) Agents trained exclusively with resilience-aligned rewards. (b) Agents trained with hybrid rewards.

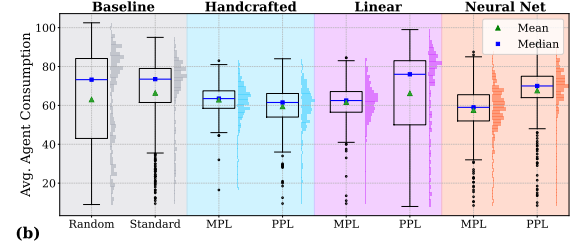
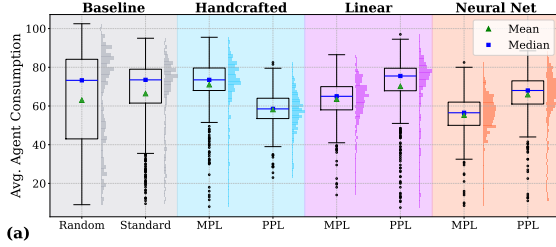


Figure A.3: Average total apple consumption per episode over 500 simulations, aggregated across both agents. (a) Agents trained exclusively with resilience-aligned rewards. (b) Agents trained with hybrid rewards.

- ϕ_4 (**Proximity Difference**): Absolute difference between the nearest-apple distances of both agents, i.e., $|\phi_2 - \phi_3|$. This feature quantifying potential asymmetries in immediate access to resources.
- ϕ_5 (**Local Apple Density (Agent 1)**): Number of apples located in the 3×3 neighborhood around Agent 1, measuring localized resource density.
- ϕ_6 (**Local Apple Density (Agent 2)**): Number of apples in the 3×3 neighborhood around Agent 2.

These features were selected to balance spatial information, resource availability, and local context, allowing the reward function to encode both global and local incentives relevant to cooperative behavior.

A.6.2 Linear Parametrization

In this case, the reward function is a linear model over the full flattened state vector, which includes agent positions and the status of each grid cell. The state is the vector $s = [x_1, x_2, y_1, y_2, m_0, m_1, \dots, m_{63}]$, where (x_1, x_2) and (y_1, y_2) denote the positions of Agent 1 and Agent 2, and $m_i \in \{0, 1\}$ indicates the presence of an apple at cell i in the 8×8 grid. This parametrization does not rely on feature engineering and instead learns directly from raw inputs. Although less interpretable than handcrafted models, it enables broader generalization by considering all available state information in a uniform representation.

Figure A.4 presents an illustrative example of the handcrafted features extracted from a specific grid configuration. In this case:

- $\phi_1 = 9$ corresponds to the total number of apples (in red) present on the grid.
- $\phi_2 = 1$ represents the distance from Agent 1 (top-left, purple) to its nearest apple, directly below.
- $\phi_3 = 0$ denotes the distance from Agent 2 (bottom-right, red) to its closest apple, also directly above.
- $\phi_4 = |\phi_2 - \phi_3| = |1 - 0| = 1$ captures the absolute difference in proximity to the nearest apples. In this case, Agent 2 is directly on top of an apple while Agent 1 is one cell away, indicating a mild asymmetry in immediate access to resources.

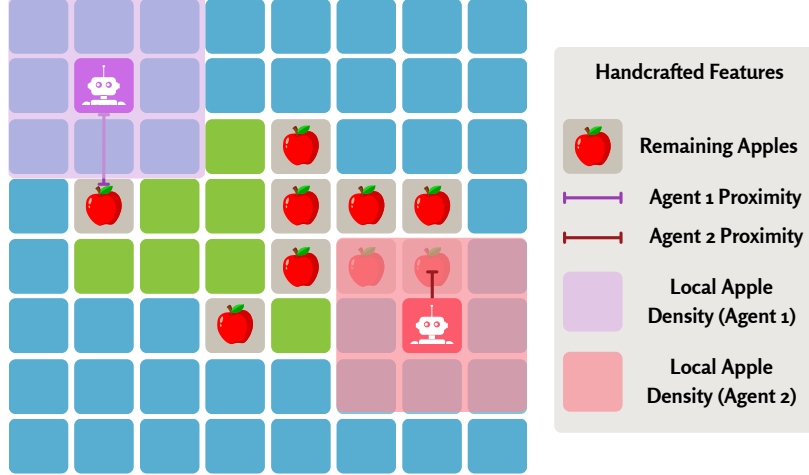


Figure A.4: Diagram of an example of handcrafted features used in the reward function. Agents (purple and red) estimate their proximity to the nearest apple, and the number of apples in their local 3×3 neighborhood.

- $\phi_5 = 0$ is the number of apples within the 3×3 neighborhood centered around Agent 1.
- $\phi_6 = 2$ is the number of apples surrounding Agent 2 in its 3×3 area.

The value of ϕ_4 is particularly informative as it reflects potential asymmetries in resource access. A low value suggests similar conditions for both agents, whereas a higher value would indicate that one agent has a significant advantage over the other in terms of proximity to resources. This feature can serve as a proxy for fairness or potential conflict over shared resources.

A.6.3 Neural Network Parametrization

The neural reward function is implemented as a fully connected feedforward neural network with the following architecture:

- Input layer of size equal to the state vector dimension (`state_dim`), which includes both agent positions and the full flattened grid.
- For **Margin-based Preference Learning**, the network uses a hidden layer with **32 units** and ReLU activation.
- For **Probabilistic Preference Learning**, the network uses a hidden layer with **64 units** and ReLU activation.
- Output layer with a single unit, representing the scalar reward.

The network is optimized using the Adam optimizer with a learning rate of 0.001. The loss function depends on the preference learning method (Margin-based Preference Learning or Probabilistic Preference Learning). This architecture was selected to provide a balance between expressive power and training efficiency, enabling the model to learn non-linear reward functions directly from raw environmental states without requiring handcrafted features.

A.7 Details on Trajectory Ranking Process

To construct the ranked dataset used for reward inference, we generated a set of **400 trajectories** using agents acting under a random policy. Each trajectory consists of **1000 timesteps**, with a **disruption event triggered at timestep 500** following the protocol described in Section A.3.

Each trajectory was evaluated using the cooperative resilience metrics detail in the main paper. These metrics include:

- **Accumulated Apple Consumption (Agent 1):** total number of apples consumed by Agent 1 over the episode.
- **Accumulated Apple Consumption (Agent 2):** same metric for Agent 2.
- **Apples Alive:** total number of apples that remained alive in the environment at the end of the episode, reflecting sustainability.
- **Equality Index:** a fairness measure indicating how evenly the apples were distributed between both agents.
- **Collective Hunger Index:** a system-level measure aggregating the number of timesteps both agents went without eating.

These trajectories were ranked according to their **cooperative resilience score**, a scalar metric combining the previous indicators to quantify the ability to anticipate, prepare for, resist, recover from, and transform in the face of disruptive events that threaten the system joint welfare.

Figure A.5 shows in (a) the distribution of cooperative resilience scores across the 400 trajectories, using a boxplot. Additionally, Figure A.5 present in (b) the distributions of final apple consumption for each agent.

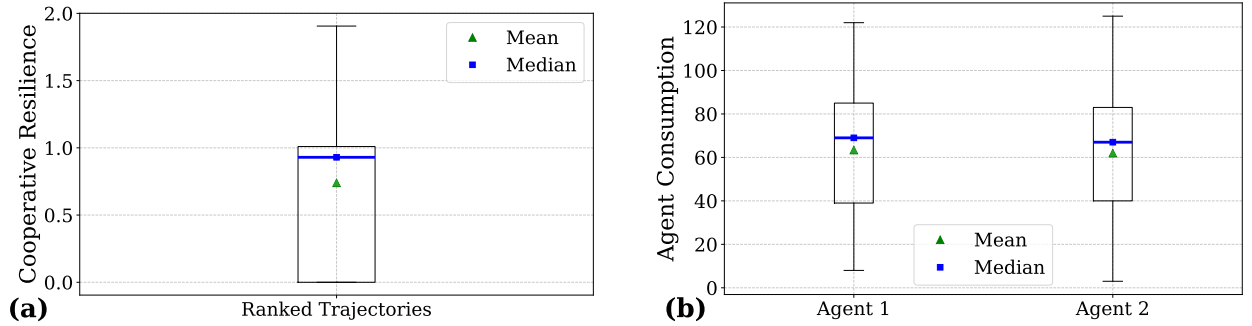


Figure A.5: Distribution of metrics in ranked trajectories. (a) Distribution of cooperative resilience scores. (b) Distribution of agent consumption.

For the 16×16 grid environment with 3 apple trees and 4 agents, we generated 400 trajectories under a random policy. Each trajectory spans 1000 timesteps, with a disruptive event consisting of apple elimination introduced at timestep 200. As in the previous environment, we used accumulated apple consumption by both agents, the number of apples remaining in the simulation, the equality index, and the collective hunger index to measure resilience. Figure A.6 in (a) shows the distribution of cooperative resilience scores across the 400 trajectories using a boxplot. In Figure A.6 panel (b) we present the cumulative apple consumption of both agents.

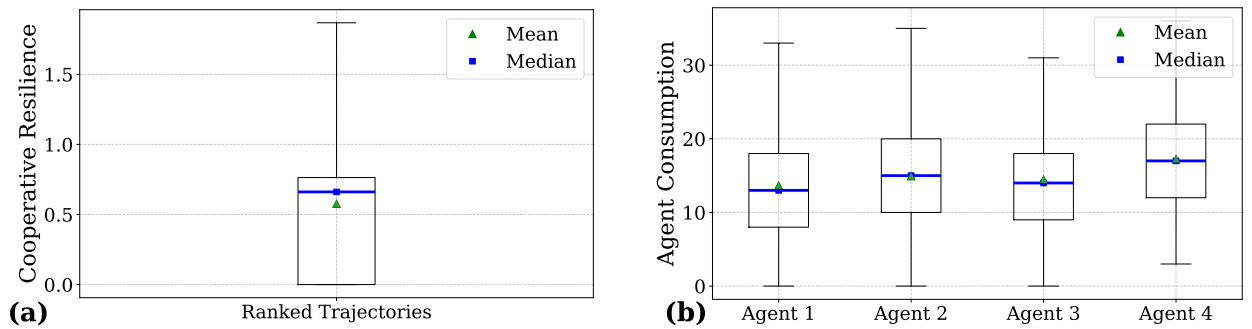


Figure A.6: Distribution of metrics in ranked trajectories in 16×16 environment. (a) Distribution of cooperative resilience scores. (b) Distribution of agent consumption.

A.8 Statistical Significance Tests

A.8.1 Evaluation Protocol

The system evaluation was conducted under the extended disruption protocol described in Appendix A.3. Across 500 test episodes, we assessed three metrics: cooperative resilience, average cumulative agent consumption, and episode length. For each metric, we performed pairwise statistical comparisons between *hybrid strategy* and the baseline methods: Random policy, PPO with the standard reward scheme, and QMIX.

Statistical significance was tested using the Mann–Whitney U test, a non-parametric test suitable for comparing two independent samples. To control for multiple hypotheses, we applied the Benjamini–Hochberg procedure (FDR, $\alpha = 0.05$) within each family of comparisons (per metric). For completeness, Bonferroni-corrected p -values are also reported. The detailed results are summarized in Tables A.7, A.8, and A.9 for cooperative resilience, average agent consumption, and episode length, respectively. In the tables, stars indicate significance levels (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

The statistical analysis confirms that *hybrid strategy* significantly outperforms both the random policy baseline and PPO in terms of cooperative resilience ($p < 10^{-60}$ after corrections). No significant difference is observed against QMIX, suggesting that both methods capture cooperative patterns of similar strength. The results for average agent consumption reinforce this finding: *hybrid strategy* significantly outperforms all three baselines, including QMIX ($p < 10^{-80}$ after corrections). Finally, the results for episode length confirm that *hybrid strategy* produces significantly longer episodes than all baselines ($p < 10^{-50}$ after corrections).

Table A.7: Pairwise comparisons of cooperative resilience. Reported p -values from Mann–Whitney U tests, adjusted within each family of three tests using Benjamini–Hochberg (BH) and Bonferroni.

Comparison	p_{raw}	p_{BH}	$p_{\text{Bonf.}}$	Sig.
<i>Hybrid strategy</i> vs Random Policy	3.53×10^{-96}	1.06×10^{-95}	1.06×10^{-95}	***
<i>Hybrid strategy</i> vs PPO (reward +1/0)	1.38×10^{-62}	2.07×10^{-62}	4.14×10^{-62}	***
<i>Hybrid strategy</i> vs QMIX (reward +1/0)	1.40×10^{-1}	1.40×10^{-1}	4.21×10^{-1}	n.s.

Table A.8: Pairwise comparisons of Agent Consumption. Reported p -values from Mann–Whitney U tests, adjusted within each family of three tests using Benjamini–Hochberg (BH) and Bonferroni.

Comparison	p_{raw}	p_{BH}	$p_{\text{Bonf.}}$	Sig.
<i>Hybrid strategy</i> vs Random Policy	6.95×10^{-85}	1.04×10^{-84}	2.08×10^{-84}	***
<i>Hybrid strategy</i> vs PPO (reward +1/0)	6.03×10^{-18}	6.03×10^{-18}	1.81×10^{-17}	***
<i>Hybrid strategy</i> vs QMIX (reward +10/0)	7.52×10^{-89}	2.26×10^{-88}	2.26×10^{-88}	***

A.8.2 Four Agents Environment

Evaluation in a more complex multi-agent setting was conducted in a 16×16 grid-world with four agents and three apple trees (see Figure A.7). In this environment, each tree disappears permanently once all surrounding apples are harvested, introducing a localized resource depletion mechanism and stronger interdependencies among agents. Moreover, resources can only regenerate up to a much lower threshold, as determined by the regeneration rule in Equation 1, where the parameter τ is fixed at 16. This implies that, regardless of the initial distribution across multiple trees, regrowth cannot exceed the equivalent of a single tree. This design increases both the spatial complexity and the consequences of overexploitation, thereby raising the challenge of sustaining cooperative resilience.

The results of 50 episodes are summarized in Figure A.8 using boxplots for cooperative resilience metrics, average final agent consumption, episode length, and bar plot for the number of episodes in which the last apple was taken. As shown, the *hybrid strategy* achieves higher values in consumption and episode length, while reducing the frequency of last-apple events associated with social dilemma failures.

To further assess the robustness of our method, we performed Mann–Whitney U tests for each metric, along with Benjamini–Hochberg and Bonferroni corrections. The detailed results are reported in Tables

Table A.9: Pairwise comparisons of Episode Length. Reported p -values from Mann–Whitney U tests, adjusted within each family of three tests using Benjamini–Hochberg (BH) and Bonferroni.

Comparison	p_{raw}	p_{BH}	$p_{\text{Bonf.}}$	Sig.
<i>Hybrid strategy</i> vs Random Policy	2.15×10^{-135}	6.45×10^{-135}	6.45×10^{-135}	***
<i>Hybrid strategy</i> vs PPO (reward +1/0)	4.97×10^{-84}	7.46×10^{-84}	1.49×10^{-83}	***
<i>Hybrid strategy</i> vs QMIX (reward +10/0)	9.80×10^{-56}	9.80×10^{-56}	2.94×10^{-55}	***

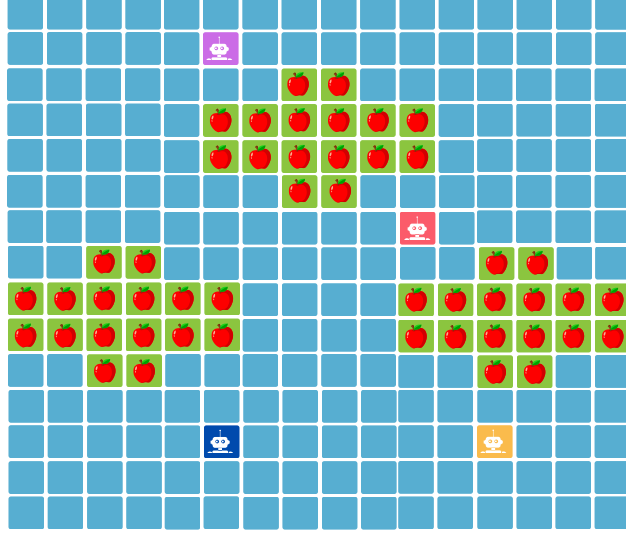


Figure A.7: Extended 16×16 grid-world environment with four agents and three apple trees. Each tree permanently disappears once all surrounding apples are harvested, introducing localized resource depletion and stronger interdependencies among agents.

A.10, A.11, and A.12 for cooperative resilience, agent consumption, and episode length, respectively. In the Tables stars indicate significance levels (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). In the Table A.10 the statistical analysis confirms that *hybrid strategy* significantly outperforms both the random policy baseline and standard PPO in terms of cooperative resilience. While the strongest effect is observed against the random baseline ($p < 10^{-9}$), the improvement over PPO also reaches statistical significance ($p < 0.05$ after corrections), indicating that resilience-based rewards provide a measurable advantage in guiding cooperative behavior. The results for average agent consumption, reported in Table A.11, reinforce the previous finding: *hybrid strategy* significantly outperforms both the random baseline and PPO ($p < 0.001$ after all corrections).

Table A.10: Pairwise comparisons of cooperative resilience. Reported p -values from Mann–Whitney U tests, adjusted within each family of three tests using Benjamini–Hochberg (BH) and Bonferroni. For 16×16 environment.

Comparison	p_{raw}	p_{BH}	$p_{\text{Bonf.}}$	Sig.
<i>Hybrid strategy</i> vs Random Policy	2.77×10^{-10}	5.53×10^{-10}	5.53×10^{-10}	***
<i>Hybrid strategy</i> vs PPO (reward +1/0)	2.27×10^{-2}	2.27×10^{-2}	4.54×10^{-2}	*

Moreover, the results for episode length, shown in Table A.12, confirm that *hybrid strategy* produces significantly longer episodes than both the random baseline and PPO ($p < 0.001$ after corrections). Together with the resilience and consumption results, this indicates that resilience-based rewards lead to more sustainable system dynamics, extending the survivability of the environment while maintaining higher levels

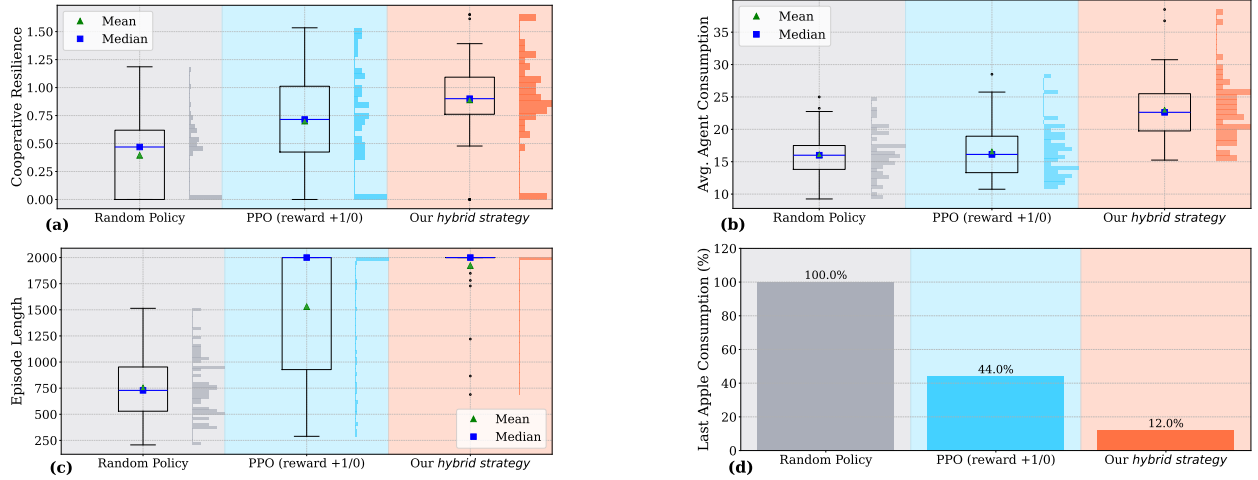


Figure A.8: Performance metrics over 50 episodes in the extended 16×16 environment with four agents and three apple trees. (a) Cooperative resilience. (b) Average total apple consumption per episode across both agents. (c) Episode length. (d) Last-apple consumption frequency, indicating the occurrence of social dilemma failures.

Table A.11: Pairwise comparisons of Average Agent Consumption. Reported p -values from Mann–Whitney U tests, adjusted within each family of three tests using Benjamini–Hochberg (BH) and Bonferroni. For 16×16 environment.

Comparison	p_{raw}	p_{BH}	$p_{\text{Bonf.}}$	Sig.
<i>Hybrid strategy</i> vs Random Policy	3.20×10^{-11}	6.39×10^{-11}	6.39×10^{-11}	***
<i>Hybrid strategy</i> vs PPO (reward +1/0)	4.00×10^{-10}	4.00×10^{-10}	8.00×10^{-10}	***

of cooperative performance.

Table A.12: Pairwise comparisons of Episode Length. Reported p -values from Mann–Whitney U tests, adjusted within each family of three tests using Benjamini–Hochberg (BH) and Bonferroni. For 16×16 environment.

Comparison	p_{raw}	p_{BH}	$p_{\text{Bonf.}}$	Sig.
<i>Hybrid strategy</i> vs Random Policy	4.43×10^{-18}	8.87×10^{-18}	8.87×10^{-18}	***
<i>Hybrid strategy</i> vs PPO (reward +1/0)	1.72×10^{-4}	1.72×10^{-4}	3.43×10^{-4}	***

B Interpretation of Learned Weights

In this section, we analyze the weights learned by the best-performing reward inference configuration, which relied on hand-crafted features, a margin-based preference learning approach, and a sampling mixture strategy (MPL-M1). This configuration, referred to as *our hybrid strategy* in the main text, achieved the highest cooperative resilience and lowest incidence of selfish behavior, also offers an interpretable structure by design.

Unlike state-based or neural parameterizations, the handcrafted reward function is explicitly defined over a set of six domain-specific features (see Section A.6.1), allowing us to directly trace the contribution of each factor to the learned incentive signal. This transparency provides insights into how the model internalizes resilience, promoting behaviors, and aligns them with measurable aspects of the environment. The learned weights thus act as a diagnostic lens, revealing which characteristics of the system were the most influential in shaping cooperative and sustainable agent behavior.

Table B.1: Interpretation of learned weights for the best handcrafted reward configuration.

Feature	Description	Learned Weight	Interpretation
ϕ_1	Remaining Apples (total apples on the grid)	w_1	Positive weight $\rightarrow$ agents are incentivized to preserve resources. Encourages sustainability. Negative weight $\rightarrow$ preference for depleted environments (possible artifact of training dynamics).
ϕ_2	Agent 1 Proximity (distance to nearest apple)	w_2	Positive weight $\rightarrow$ avoidance behavior, potentially to reduce competition. Negative weight $\rightarrow$ encourages closer proximity to resources, signaling proactive behavior.
ϕ_3	Agent 2 Proximity (distance to nearest apple)	w_3	Similar to ϕ_2 . If the sign differs from w_2 , it may suggest role specialization or asymmetric coordination between agents.
ϕ_4	Proximity difference (distance inequality)	w_4	Positive weight $\rightarrow$ penalizes asymmetric access to nearby apples. Encourages fairness and spatial balance between agents. Negative weight $\rightarrow$ negative weights may signal tolerance for emergent role differentiation.
ϕ_5	Local Apple Density (Agent 1)	w_5	Positive weight $\rightarrow$ encourages Agent 1 to operate in resource-rich areas. If significantly larger than w_6 , may indicate structural bias.
ϕ_6	Local Apple Density (Agent 2)	w_6	Same interpretation as ϕ_5 , but for Agent 2. Comparing w_5 and w_6 helps diagnose symmetry or potential role division.

Figure B.1 illustrates the learned weights associated with the handcrafted reward function in the best-performing configuration. The relative magnitudes of the weights provide interpretable signals about the emergent behaviors favored by the inferred reward.

Among the six features, the highest weights were assigned to *Agent 2 Proximity* (ϕ_3), *Local Apple Density for Agent 1* (ϕ_5), and *Agent 1 Proximity* (ϕ_2). Since proximity features encode the distance to the nearest apple, their high positive weights indicate that agents were encouraged to maintain a degree of distance from individual resources, possibly to reduce direct competition or avoid overexploitation. In contrast, the high weights on local density suggest that agents were incentivized to remain within resource-rich areas, without necessarily needing immediate access to a single apple. This combination promotes spatial behaviors in which agents position themselves strategically, away from direct contention, but still within productive regions.

The feature *Remaining Apples* (ϕ_1) received a smaller positive weight, suggesting that system-level sustainability was modestly rewarded but not heavily prioritized. *Proximity Difference* (ϕ_4) captures the absolute difference between the distances of both agents to their nearest apple, quantifying potential asymmetries in immediate access to resources. A negative weight on ϕ_4 might, at first glance, suggest that the model favors *inequality* in access to resources. However, in our setting, this outcome is more plausibly interpreted as the emergence of complementary behaviors, where agents adopt specialized roles. For instance, one agent may consistently position itself closer to resources to prioritize collection, while the other remains farther away, potentially exploring. These spatial asymmetries are not inherently unfair, in fact, they can reflect strategies that enhance overall system performance. A detailed interpretation of each feature and its associated weight is presented in Table B.1.

Interestingly, agents trained under this reward displayed specialized spatial behaviors. One agent actively explored broader areas of the grid, while the other remained anchored along the right boundary, consistently harvesting nearby resources with limited movement. This emergent division of roles can be interpreted

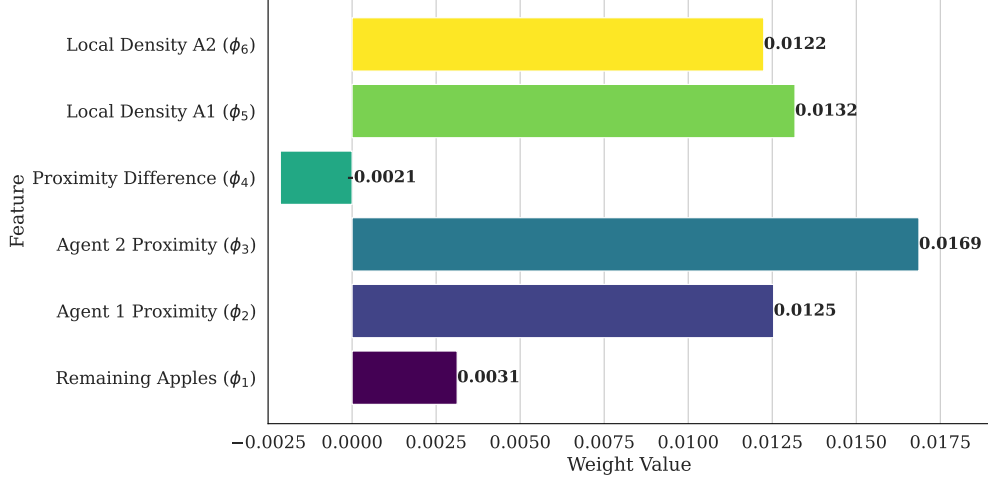


Figure B.1: Learned Weights with the handcrafted reward function in *hybrid strategy*.

as a consequence of the learned incentives, which balance the need to exploit resource-rich zones while avoiding redundant overlap. Such specialization illustrates how the reward structure supports complementary strategies. See supplementary material for a video example of trained agent behavior <https://drive.google.com/file/d/1li7roG6AW0nxAOeBwD1194SpoLFFeyw2/view?usp=sharing>.

C Additional Spatial Behavior Visualizations

In this appendix, we provide additional visualizations of agent positions. Figure C.1 shows disaggregated frequency maps for each agent, while Figure C.2 presents circle-based maps where the size of each circle indicates how frequently a location was visited. The latter makes it easy to see the specialized behavior in the *hybrid strategy*, where one agent stays near the boundary and the other explores more widely.

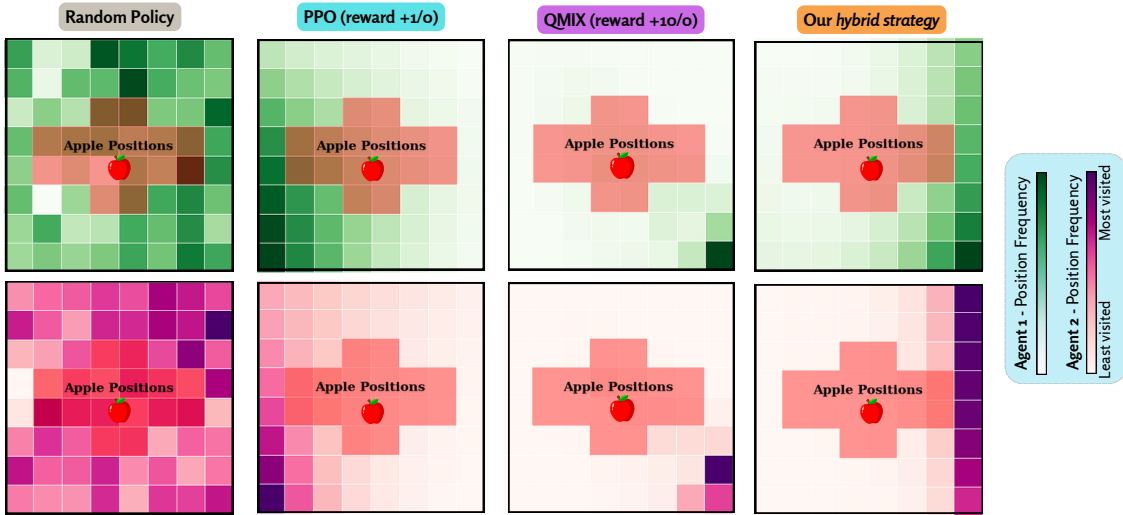


Figure C.1: Individual position frequency maps for Agent 1 (green) and Agent 2 (purple) under four training configurations: (i) random policy, (ii) PPO with standard rewards, (iii) QMIX, and (iv) our *hybrid strategy*. Each heatmap depicts the spatial visitation density over 500 evaluation episodes in an 8×8 grid, with apple locations marked in red. Agents were randomly initialized at the start of each episode and evaluated under the same protocol with three disruption events.

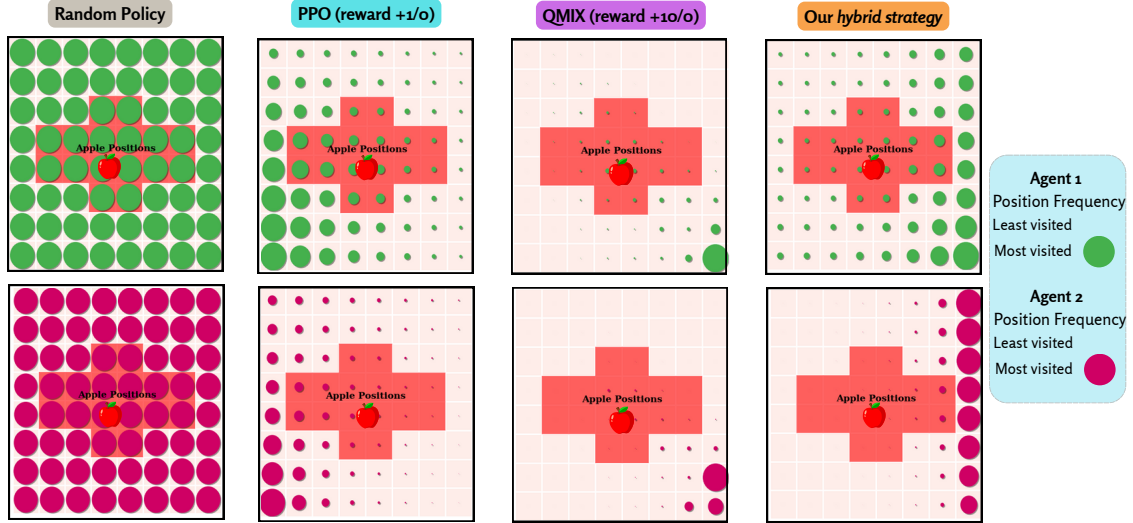


Figure C.2: Circle-based position maps for Agent 1 (green) and Agent 2 (purple) under the four training configurations: (i) random policy, (ii) PPO with standard rewards, (iii) QMIX, and (iv) our *hybrid strategy*. Circle size is proportional to the visit frequency at each grid location, and very low-frequency visits are omitted for clarity.

D Societal Impact Considerations

This work contributes to the design multi-agent AI systems that are more resilient to disruptions in shared-resource environments. By focusing on reward learning aligned with collective welfare and recovery, the framework promotes agent behaviors that go beyond individual optimization, encouraging fairness, sustainability, and adaptability.

Positive Impacts. The proposed approach may contribute to the development of multi-agent AI systems capable of operating in dynamic environments such as disaster response, resource management, and multi-robot coordination, where resilience in support of collective well-being is essential. Reward functions inferred from resilient behaviors can act as proxies for socially desirable values, including equity, mutual support, and long-term sustainability—even in systems composed of both artificial and non-artificial (e.g., human) agents.

Negative Impacts and Limitations If misused or deployed without oversight, cooperative reward functions could be manipulated to favor specific agents or reinforce harmful coordination patterns (e.g., excluding certain agents from resources). Furthermore, our method relies on trajectory ranking heuristics that may not fully capture human values. In real-world deployment, this may raise risks related to fairness, representation, or trust.

Mitigation Measures. We encourage researchers to incorporate human-in-the-loop oversight when defining ranking criteria or interpreting resilience-based rewards, especially in domains with ethical, social, or legal implications. Transparency in how reward functions are learned and applied is critical for accountability and alignment with stakeholder values.

While this work does not involve direct deployment or the use of sensitive data, its broader applicability to cooperative AI systems calls for careful consideration of potential misuse—particularly in socio-technical contexts, high-risk domains involving human participants, or settings where agent decisions directly impact real-world resource management.

E Limitations

Our results demonstrate that resilience-aligned rewards improve cooperative behavior in mixed-motive settings. However, the experiments are conducted in a simplified and fully observable environment with discrete actions and known dynamics. We additionally introduced a more complex setting with a larger search space, more agents, and apple trees with limited regeneration, yet the conditions remain constrained to grid-world environments. Thus, these experiments may not fully capture the complexity of real-world multi-agent systems, where agents often face partial observability, larger and continuous state spaces, and dynamic interaction topologies. Additionally, the resilience metric used to rank trajectories is manually designed for the specific task, which may limit its applicability across domains.

F Reproducibility Assets

In addition to the main manuscript, we provide as supplementary material a Code Ocean capsule containing the source code and experimental configurations used in this study, available at: <https://codeocean.com/capsule/4898801/tree/v1>. The capsule includes detailed instructions for reproducing the experiments through the accompanying `README.md` file. For completeness, a compressed `.zip` file containing the source code is also included with this submission.